
PRIOR AUTHORIZATION TRIAGE & POLICY COMPANION
Output Explainability Demo (LIVE NEON POSTGRESQL)
LLM Mode : LIVE - qwen.qwen3-next-80b-a3b via BEDROCK
Repository Mode: LIVE NEON POSTGRESQL

APPROVE - Covered dx + J5 state (TX) [LCD Path]

REQUEST

Procedure Code : 64483
Diagnosis : M54.16
State : TX
Patient Age : 55
Clinical Notes : Patient presents with lumbar radiculopathy confirmed on MRI. Conservative therapy was tried for 8 weeks without relief.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED
Jurisdiction : TX -> MATCHED
LCD : 36920 Epidural Steroid Injections for Pain Management
Article : 56681

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

CODE MATCHING

HCPCS 64483 -> Article 56681 -> MATCHED
ICD-10 M54.16 -> Article 56681 -> COVERED

POLICY CRITERIA

C1 [SQL] ARTICLE-56681-HCPCS

Requirement : The requested procedure must be an applicable service under the Article.

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : Article 56681 contains HCPCS 64483 in its coverage list.

Patient Ev. : Submitted HCPCS: 64483

Explanation : Evaluated as SATISFIED by SQL.

C2 [SQL] ARTICLE-56681-ICD10-M54.16

Requirement : The patient's diagnosis must be an eligible diagnosis under the Article.

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : Diagnosis M54.16 is present in the Article's covered ICD-10 data.

Patient Ev. : Submitted ICD-10: M54.16

Explanation : Evaluated as SATISFIED by SQL.

EVIDENCE FUSION

Criteria SATISFIED : 2

Criteria NOT_SATISFIED : 0

Criteria UNKNOWN : 0

Evidence Fusion Result : COVERED

FINAL DECISION

[APPROVE] Decision : APPROVE

Reason : All applicable policy criteria were satisfied. The submitted procedure and diagnosis codes are covered under the applicable Article. [ARTICLE_CRITERIA_SATISFIED]

Reason Codes: ARTICLE_CRITERIA_SATISFIED

Decision Basis:

All mandatory policy criteria were satisfied.

* ARTICLE-56681-HCPCS: SATISFIED

* ARTICLE-56681-ICD10-M54.16: SATISFIED

Evidence Fusion: COVERED

DecisionEngine: COVERED -> APPROVE

PEND - Explicitly non-covered diagnosis [LCD 39662 Path]

REQUEST

Procedure Code : 20552

Diagnosis : M25.50

State : TX

Patient Age : 40

Clinical Notes : Routine consultation for general unspecified joint pain without muscle trigger points.

POLICY IDENTIFICATION

NCD : 373

Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

POLICY CRITERIA

C1 [SQL] NCD-373-HCPCS

Requirement : The requested procedure must not be explicitly excluded by NCD 373.

Type : STRUCTURED

Result : NOT_SATISFIED

Policy Ev. : NCD 373 explicitly excludes HCPCS 20552.

Patient Ev. : Submitted HCPCS: 20552

Explanation : Procedure 20552 is listed in NCD 373 which EXCLUDES coverage. Criterion NOT_SATISFIED by deterministic SQL

check.

EVIDENCE FUSION

Criteria SATISFIED : 0
Criteria NOT_SATISFIED : 1
Criteria UNKNOWN : 0
Evidence Fusion Result : EXCLUDED

FINAL DECISION

[PEND] Decision : PEND
Reason : The submitted procedure is explicitly excluded by an applicable National Coverage Determination (NCD). The request is pending for manual review. [NCD_EXCLUDES_PROCEDURE]
Reason Codes: NCD_EXCLUDES_PROCEDURE

Decision Basis:
The request was pending because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.
* NCD-373-HCPCS: NOT_SATISFIED
Evidence Fusion: EXCLUDED
DecisionEngine: EXCLUDED -> PEND

[94m2026-08-16 13:19:06 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "Screening for Hepatitis C Virus (HCV) in adults", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "HCV_screening_indication", [0m
[36m "description": "Patient is an adult requiring HCV screening as per policy" [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 526 | output_tokens: 65 | total_tokens: 591 | latency: 995ms [0m

[94m2026-08-16 13:19:09 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
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[36m "Patient was born between 1945 and 1965 with history of blood transfusion." [0m
[36m], [0m
[36m "contradicting_evidence": [], [0m
[36m "missing_evidence": [] [0m
[36m } [0m
[93m [USAGE] input_tokens: 751 | output_tokens: 75 | total_tokens: 826 | latency: 1372ms [0m

[94m2026-08-16 13:19:11 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "SATISFIED", [0m
[36m "evidence_cited": [[0m
[36m "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.", [0m
[36m "Patient was born between 1945 and 1965 with history of blood transfusion." [0m

[36m], [0m
[36m "explanation": "The patient meets both age-based (1945\u20131965 birth cohort) and high-risk (history of blood transfusion) criteria for HCV screening per standard guidelines." [0m
[36m } [0m
[93m [USAGE] input_tokens: 373 | output_tokens: 112 | total_tokens: 485 | latency: 996ms [0m

[94m2026-08-16 13:19:13 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "HCV screening is covered for adults at high risk for HCV infection", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "high_risk_status", [0m
[36m "description": "Patient must be an adult identified as being at high risk for HCV infection" [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 522 | output_tokens: 68 | total_tokens: 590 | latency: 777ms [0m

[94m2026-08-16 13:19:15 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
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[36m "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.", [0m
[36m "Patient was born between 1945 and 1965 with history of blood transfusion." [0m
[36m], [0m
[36m "contradicting_evidence": [], [0m
[36m "missing_evidence": [] [0m
[36m } [0m
[93m [USAGE] input_tokens: 754 | output_tokens: 75 | total_tokens: 829 | latency: 1233ms [0m

[94m2026-08-16 13:19:18 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
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[36m "result": "SATISFIED", [0m
[36m "evidence_cited": [[0m
[36m "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.", [0m
[36m "Patient was born between 1945 and 1965 with history of blood transfusion." [0m
[36m], [0m
[36m "explanation": "The patient is an adult with known HCV risk factors (birth cohort 1945\u20131965 and history of blood transfusion), satisfying the policy requirement for high-risk HCV screening." [0m
[36m } [0m
[93m [USAGE] input_tokens: 374 | output_tokens: 116 | total_tokens: 490 | latency: 1664ms [0m

[94m2026-08-16 13:19:20 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "Coverage is for adults who do not meet the high risk definition but were born between 1945 and 1965", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "birth_year_range", [0m
[36m "description": "Patient must have been born between 1945 and 1965" [0m
[36m], [0m

[36m { [0m
[36m "category": "not_high_risk", [0m
[36m "description": "Patient must not meet the high risk definition stated above" [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 560 | output_tokens: 112 | total_tokens: 672 | latency: 1202ms [0m

[94m2026-08-16 13:19:23 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "supporting_evidence": [[0m
[36m "Patient was born between 1945 and 1965" [0m
[36m], [0m
[36m "contradicting_evidence": [[0m
[36m "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient" [0m
[36m], [0m
[36m "missing_evidence": [[0m
[36m "Patient must not meet the high risk definition stated above" [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 787 | output_tokens: 74 | total_tokens: 861 | latency: 1126ms [0m

[94m2026-08-16 13:19:25 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "UNKNOWN", [0m
[36m "evidence_cited": [[0m
[36m "Patient was born between 1945 and 1965" [0m
[36m], [0m
[36m "explanation": "While the birth year range is satisfied, there is no evidence confirming the patient does not meet the high-risk definition, which is required." [0m
[36m } [0m
[93m [USAGE] input_tokens: 422 | output_tokens: 67 | total_tokens: 489 | latency: 1036ms [0m

[94m2026-08-16 13:19:29 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "Cover screening for HCV in adults at high risk for HCV infection and one-time screening for adults born between 1945 and 1965, if the service is recommended with a grade A or B by the USPSTF and meets statutory requirements for preventive services.", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "USPSTF_recommendation_grade", [0m
[36m "description": "Service must be recommended with a grade of A or B by the United States Preventive Services Task Force (USPSTF)" [0m
[36m], [0m
[36m { [0m
[36m "category": "screening_population_criteria", [0m
[36m "description": "Patient must be either an adult at high risk for HCV infection OR an adult born between 1945 and 1965" [0m
[36m], [0m
[36m { [0m
[36m "category": "preventive_service_necessity", [0m

[36m "description": "Service must be reasonable and necessary for the prevention or early detection of illness or disability" [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 1580 | output_tokens: 212 | total_tokens: 1792 | latency: 3026ms [0m

[94m2026-08-16 13:19:32 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "supporting_evidence": [[0m
[36m "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.", [0m
[36m "Patient was born between 1945 and 1965 with history of blood transfusion." [0m
[36m], [0m
[36m "contradicting_evidence": [], [0m
[36m "missing_evidence": [[0m
[36m "USPSTF_recommendation_grade: Evidence of a USPSTF grade A or B recommendation for the service is not stated." [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 876 | output_tokens: 106 | total_tokens: 982 | latency: 1683ms [0m

[94m2026-08-16 13:19:35 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "UNKNOWN", [0m
[36m "evidence_cited": [[0m
[36m "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.", [0m
[36m "Patient was born between 1945 and 1965 with history of blood transfusion." [0m
[36m], [0m
[36m "explanation": "Although the patient meets the screening population criteria, the USPSTF recommendation grade (A or B) for
the service is not explicitly stated in the evidence, making it insufficient to confirm policy satisfaction." [0m
[36m } [0m
[93m [USAGE] input_tokens: 1049 | output_tokens: 111 | total_tokens: 1160 | latency: 1666ms [0m

[94m2026-08-16 13:19:37 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "Screening for Hepatitis B Virus (HBV) Infection", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "hbv_screening_indication", [0m
[36m "description": "Patient must be undergoing screening for Hepatitis B Virus (HBV) infection as defined by the policy" [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 526 | output_tokens: 74 | total_tokens: 600 | latency: 1247ms [0m

[94m2026-08-16 13:19:39 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "supporting_evidence": [], [0m
[36m "contradicting_evidence": [[0m
[36m "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient." [0m
[36m], [0m

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[36m  "missing_evidence": [ [0m
[36m    "Patient must be undergoing screening for Hepatitis B Virus (HBV) infection as defined by the policy" [0m
[36m  ] [0m
[36m ] [0m
[93m [USAGE] input_tokens: 760 | output_tokens: 69 | total_tokens: 829 | latency: 956ms [0m
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[94m2026-08-16 13:19:41 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m  "result": "NOT_SATISFIED", [0m
[36m  "evidence_cited": [ [0m
[36m    "Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient." [0m
[36m  ], [0m
[36m  "explanation": "The patient evidence shows screening for HCV, not HBV, which directly contradicts the policy requirement for
HBV screening." [0m
[36m ] [0m
[93m [USAGE] input_tokens: 376 | output_tokens: 74 | total_tokens: 450 | latency: 1011ms [0m
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[94m2026-08-16 13:19:43 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m  "requirement": "Coverage is provided for asymptomatic, nonpregnant adolescents and adults at high risk for HBV
infection", [0m
[36m  "required_evidence": [ [0m
[36m    { [0m
[36m      "category": "patient_status", [0m
[36m      "description": "Patient must be asymptomatic, nonpregnant, and either an adolescent or adult" [0m
[36m    ], [0m
[36m    { [0m
[36m      "category": "risk_status", [0m
[36m      "description": "Patient must be at high risk for HBV infection" [0m
[36m    ] [0m
[36m  ] [0m
[36m ] [0m
[93m [USAGE] input_tokens: 538 | output_tokens: 101 | total_tokens: 639 | latency: 1109ms [0m
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[94m2026-08-16 13:19:46 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m  "supporting_evidence": [ [0m
[36m    "asymptomatic high-risk adult patient", [0m
[36m    "Patient Age: 52", [0m
[36m    "Patient was born between 1945 and 1965 with history of blood transfusion." [0m
[36m  ], [0m
[36m  "contradicting_evidence": [], [0m
[36m  "missing_evidence": [ [0m
[36m    "Evidence that the patient is at high risk for HBV infection (the data mentions HCV screening and risk factors for HCV, not
HBV)" [0m
[36m  ] [0m
[36m ] [0m
[93m [USAGE] input_tokens: 776 | output_tokens: 108 | total_tokens: 884 | latency: 1053ms [0m
```

[94m2026-08-16 13:19:48 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "UNKNOWN", [0m
[36m "evidence_cited": [[0m
[36m "asymptomatic high-risk adult patient", [0m
[36m "Patient Age: 52", [0m
[36m "Patient was born between 1945 and 1965 with history of blood transfusion." [0m
[36m], [0m
[36m "explanation": "Although the patient is an asymptomatic, nonpregnant adult, the evidence cites HCV risk factors and does not confirm HBV-specific high-risk status as required by the policy." [0m
[36m } [0m
[93m [USAGE] input_tokens: 429 | output_tokens: 101 | total_tokens: 530 | latency: 1309ms [0m

[94m2026-08-16 13:19:51 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "Screening is covered for pregnant women, and rescreening at time of delivery is covered for those with new or continuing risk factors", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "pregnancy_status", [0m
[36m "description": "Patient must be pregnant" [0m
[36m], [0m
[36m { [0m
[36m "category": "risk_factors", [0m
[36m "description": "Patient must have new or continuing risk factors at time of delivery" [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 544 | output_tokens: 99 | total_tokens: 643 | latency: 1735ms [0m

[94m2026-08-16 13:19:53 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "supporting_evidence": [], [0m
[36m "contradicting_evidence": [], [0m
[36m "missing_evidence": [[0m
[36m "pregnancy_status: Patient must be pregnant", [0m
[36m "risk_factors: Patient must have new or continuing risk factors at time of delivery" [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 774 | output_tokens: 60 | total_tokens: 834 | latency: 954ms [0m

[94m2026-08-16 13:19:55 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "UNKNOWN", [0m
[36m "evidence_cited": [], [0m
[36m "explanation": "No patient evidence is available to confirm pregnancy status or the presence of new or continuing risk factors at time of delivery." [0m
[36m } [0m
[93m [USAGE] input_tokens: 390 | output_tokens: 47 | total_tokens: 437 | latency: 528ms [0m

REQUEST

Procedure Code : 87556

Diagnosis : Z11.59

State : TX

Patient Age : 52

Clinical Notes : Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient. Patient was born between 1945 and 1965 with history of blood transfusion.

POLICY IDENTIFICATION

NCD : 352

Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD 361

Section : title

Text : Screening for Hepatitis C Virus (HCV) in Adults

Similarity : 0.6466

Source : NCD 361

Section : indications_limitations

Text : B. Nationally Covered Indications Effective for services performed on or after June 2, 2014, CMS has determined the following: The evidence is adequate to conclude that screening for HCV, consist

Similarity : 0.5349

Source : NCD 361

Section : item_service_description

Text : A. General Hepatitis C Virus (HCV) is an infection that attacks the liver and leads to inflammation. The infection is often asymptomatic and can go undiagnosed for decades. It is difficult for th

Similarity : 0.5344

Source : NCD 369

Section : title

Text : Screening for Hepatitis B Virus (HBV) Infection

Similarity : 0.4784

Source : NCD 369

Section : indications_limitations

Text : B. Nationally Covered Indications Effective for services performed on or after September 28, 2016, CMS has determined that the evidence is sufficient to cover screening for HBV infection with the appr

Similarity : 0.4758

POLICY CRITERIA

C1 [SQL] NCD-352-HCPCS

Requirement : The requested procedure must not be explicitly excluded by NCD 352.

Type : STRUCTURED

Result : NOT_SATISFIED

Policy Ev. : NCD 352 explicitly excludes HCPCS 87556.

Patient Ev. : Submitted HCPCS: 87556

Explanation : Procedure 87556 is listed in NCD 352 which EXCLUDES coverage. Criterion NOT_SATISFIED by deterministic SQL check.

C2 [SQL] NCD-361-HCPCS

Requirement : The requested procedure must not be explicitly excluded by NCD 361.

Type : STRUCTURED

Result : NOT_SATISFIED

Policy Ev. : NCD 361 explicitly excludes HCPCS 87556.

Patient Ev. : Submitted HCPCS: 87556

Explanation : Procedure 87556 is listed in NCD 361 which EXCLUDES coverage. Criterion NOT_SATISFIED by deterministic SQL check.

C3 [SQL] NCD-365-HCPCS

Requirement : The requested procedure must not be explicitly excluded by NCD 365.

Type : STRUCTURED

Result : NOT_SATISFIED

Policy Ev. : NCD 365 explicitly excludes HCPCS 87556.

Patient Ev. : Submitted HCPCS: 87556

Explanation : Procedure 87556 is listed in NCD 365 which EXCLUDES coverage. Criterion NOT_SATISFIED by deterministic SQL check.

C4 [SQL] NCD-369-HCPCS

Requirement : The requested procedure must not be explicitly excluded by NCD 369.

Type : STRUCTURED

Result : NOT_SATISFIED

Policy Ev. : NCD 369 explicitly excludes HCPCS 87556.

Patient Ev. : Submitted HCPCS: 87556

Explanation : Procedure 87556 is listed in NCD 369 which EXCLUDES coverage. Criterion NOT_SATISFIED by deterministic SQL check.

C5 [AGENTIC_QWEN] NCD-361-Cfcee62

Requirement : Screening for Hepatitis C Virus (HCV) in Adults

Type : SEMANTIC

Result : SATISFIED

Policy Ev. : Required: HCV_screening_indication: Patient is an adult requiring HCV screening as per policy

Patient Ev. : Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.; Patient was born between 1945 and 1965 with history of blood transfusion.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Screening for Hepatitis C Virus (HCV) in Adults

Policy: NCD 361

Required Evidence:

* HCV_screening_indication: Patient is an adult requiring HCV screening as per policy

Patient Evidence:

* Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.

* Patient was born between 1945 and 1965 with history of blood transfusion.

Qwen Result: SATISFIED

Critic Result: VALIDATED

Final Result: SATISFIED

The submitted clinical documentation satisfies this semantic policy criterion. The agentic evaluation chain VALIDATED the result.

(Agentic pipeline completed in 7192ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 2531ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 0 missing evidence items in 2566ms.
[EVALUATION_AGENT] COMPLETED: Supporting evidence found (2 items). No missing evidence categories. Policy requirement may be satisfied.
[QWEN] COMPLETED: Qwen returned SATISFIED. Evidence cited: 2 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: SATISFIED. All 6 checks passed.

C6 [AGENTIC_QWEN] NCD-361-C00c478
Requirement : is covered for adults at high risk for HCV infection
Type : SEMANTIC
Result : SATISFIED
Policy Ev. : Required: high_risk_status: Patient must be an adult identified as being at high risk for HCV infection
Patient Ev. : Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.; Patient was born between 1945 and 1965 with history of blood transfusion.
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: is covered for adults at high risk for HCV infection
Policy: NCD 361

Required Evidence:
* high_risk_status: Patient must be an adult identified as being at high risk for HCV infection

Patient Evidence:
* Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.
* Patient was born between 1945 and 1965 with history of blood transfusion.

Qwen Result: SATISFIED
Critic Result: VALIDATED
Final Result: SATISFIED

The submitted clinical documentation satisfies this semantic policy criterion. The agentic evaluation chain VALIDATED the result.

(Agentic pipeline completed in 6910ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 1849ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 0 missing evidence items in 2313ms.
[EVALUATION_AGENT] COMPLETED: Supporting evidence found (2 items). No missing evidence categories. Policy requirement may be satisfied.
[QWEN] COMPLETED: Qwen returned SATISFIED. Evidence cited: 2 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: SATISFIED. All 6 checks passed.

C7 [AGENTIC_QWEN] NCD-361-C634c41
Requirement : is covered for adults who do not meet the high risk definition above, but who were born from 1945 through 1965
Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: birth_year_range: Patient must have been born between 1945 and 1965; Required: not_high_risk: Patient must not meet the high risk definition stated above

Patient Ev. : Patient was born between 1945 and 1965

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for adults who do not meet the high risk definition above, but who were born from 1945 through 1965
Policy: NCD 361

Required Evidence:
* birth_year_range: Patient must have been born between 1945 and 1965
* not_high_risk: Patient must not meet the high risk definition stated above

Patient Evidence:
* Patient was born between 1945 and 1965

Missing Evidence:
* Patient must not meet the high risk definition stated above

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 6914ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 2466ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 1 contradicting, 1 missing evidence items in 2239ms.
[EVALUATION_AGENT] COMPLETED: Contradicting evidence found (1 items). Policy requirement appears NOT satisfied.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C8 [AGENTIC_QWEN] NCD-361-Ceb1ff3
Requirement : A. General Hepatitis C Virus (HCV) is an infection that attacks the liver and leads to inflammation. The ...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: USPSTF_recommendation_grade: Service must be recommended with a grade of A or B by the United States Preventive Services Task Force (USPSTF); Required: screening_population_criteria: Patient must be either an adult at high risk for HCV infection OR an adult born between 1945 and 1965
Patient Ev. : Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.; Patient was born between 1945 and 1965 with history of blood transfusion.
Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: A. General Hepatitis C Virus (HCV) is an infection that attacks the liver and leads to inflammation. The infection is often asymptomatic and can go undiagnosed for decades. It is difficult for the human immune system to eliminate the HCV and it is a major cause of chronic liver disease. The presence of HCV in the liver initiates a response from the immune system which in turn causes inflammation. Inflammation over long periods of time (usually decades) can cause scarring, called cirrhosis. A cirrhotic liver fails to perform the normal functions of the liver which leads to liver failure. Cirrhotic livers are more prone to become cancerous and

liver failure leads to serious complications, even death. HCV is reported to be the leading cause of chronic hepatitis, cirrhosis and liver cancer and a primary indication for liver transplant in the Western World. Under §1861(ddd) of the Social Security Act (the Act), the Centers for Medicare & Medicaid Services (CMS) has the authority to add coverage of additional preventive services if certain statutory requirements are met. The regulations provide: 42 CFR §410.64 Additional preventive services (a) Medicare Part B pays for additional preventive services not described in paragraph (1) or (3) of the definition of preventive services under 42 CFR §410.2, that identify medical conditions or risk factors for individuals if the Secretary determines through the national coverage determination process (as defined in section 1869(f)(1)(B) of the Act) that these services are all of the following: (1) reasonable and necessary for the prevention or early detection of illness or disability, (2) recommended with a grade of A or B by the United States Preventive Services Task Force (USPSTF), (3) appropriate for individuals entitled to benefits under Part A or enrolled under Part B. (b) In making determinations under paragraph (a) of this section regarding the coverage of a new preventive service, the Secretary may conduct an assessment of the relation between predicted outcomes and the expenditures for such services and may take into account the results of such an assessment in making such national coverage determinations. The scope of the review for this national coverage determination (NCD) evaluated and determined if the existing body of evidence was sufficient for Medicare coverage for screening for HCV in adults at high risk for HCV infection and one-time screening for HCV infection for adults born between 1945 and 1965, which is recommended with a grade B by the USPSTF.

Policy: NCD 361

Required Evidence:

- * USPSTF_recommendation_grade: Service must be recommended with a grade of A or B by the United States Preventive Services Task Force (USPSTF)
- * screening_population_criteria: Patient must be either an adult at high risk for HCV infection OR an adult born between 1945 and 1965
- * preventive_service_necessity: Service must be reasonable and necessary for the prevention or early detection of illness or disability

Patient Evidence:

- * Screening for Hepatitis C Virus (HCV) in asymptomatic high-risk adult patient.
- * Patient was born between 1945 and 1965 with history of blood transfusion.

Missing Evidence:

- * USPSTF_recommendation_grade: Evidence of a USPSTF grade A or B recommendation for the service is not stated.

Qwen Result: UNKNOWN

Critic Result: VALIDATED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 9732ms)

Agent Trace:

- [POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 4202ms.
- [CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 1 missing evidence items in 2797ms.
- [EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 1 required evidence categories are missing. Evidence is insufficient for a definitive conclusion.
- [QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 2 items.
- [CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C9 [AGENTIC_QWEN] NCD-369-Ca67265

Requirement : Screening for Hepatitis B Virus (HBV) Infection

Type : SEMANTIC

Result : NOT_SATISFIED

Policy Ev. : Required: hbv_screening_indication: Patient must be undergoing screening for Hepatitis B Virus (HBV) infection as defined by the policy

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Screening for Hepatitis B Virus (HBV) Infection
Policy: NCD 369

Required Evidence:
* hbv_screening_indication: Patient must be undergoing screening for Hepatitis B Virus (HBV) infection as defined by the policy

Patient Evidence:
* (No supporting evidence found in request)

Missing Evidence:
* Patient must be undergoing screening for Hepatitis B Virus (HBV) infection as defined by the policy

Qwen Result: NOT_SATISFIED
Critic Result: VALIDATED
Final Result: NOT_SATISFIED

The submitted clinical documentation does not satisfy this semantic policy criterion based on available evidence.

(Agentic pipeline completed in 6533ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 2304ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 0 supporting, 1 contradicting, 1 missing evidence items in 2068ms.
[EVALUATION_AGENT] COMPLETED: Contradicting evidence found (1 items). Policy requirement appears NOT satisfied.
[QWEN] COMPLETED: Qwen returned NOT_SATISFIED. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: NOT_SATISFIED. All 6 checks passed.

C10 [AGENTIC_QWEN] NCD-369-C3acd4a
Requirement : is covered for asymptomatic, nonpregnant adolescents and adults at high risk for HBV infection
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: patient_status: Patient must be asymptomatic, nonpregnant, and either an adolescent or adult; Required: risk_status: Patient must be at high risk for HBV infection
Patient Ev. : asymptomatic high-risk adult patient; Patient Age: 52; Patient was born between 1945 and 1965 with history of blood transfusion.
Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: is covered for asymptomatic, nonpregnant adolescents and adults at high risk for HBV infection
Policy: NCD 369

Required Evidence:
* patient_status: Patient must be asymptomatic, nonpregnant, and either an adolescent or adult
* risk_status: Patient must be at high risk for HBV infection

Patient Evidence:
* asymptomatic high-risk adult patient
* Patient Age: 52

* Patient was born between 1945 and 1965 with history of blood transfusion.

Missing Evidence:

* Evidence that the patient is at high risk for HBV infection (the data mentions HCV screening and risk factors for HCV, not HBV)

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 6844ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 2194ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 3 supporting, 0 contradicting, 1 missing evidence items in 2251ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 3 supporting items found, but 1 required evidence categories are missing.
Evidence is insufficient for a definitive conclusion.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 3 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C11 [AGENTIC_QWEN] NCD-369-C9731ee
Requirement : is covered for pregnant women and then rescreening at time of delivery for those with new or continuing risk f...
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: pregnancy_status: Patient must be pregnant; Required: risk_factors: Patient must have new or continuing risk factors at time of delivery
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: is covered for pregnant women and then rescreening at time of delivery for those with new or continuing risk factors
Policy: NCD 369

Required Evidence:

* pregnancy_status: Patient must be pregnant
* risk_factors: Patient must have new or continuing risk factors at time of delivery

Patient Evidence:

* (No supporting evidence found in request)

Missing Evidence:

* pregnancy_status: Patient must be pregnant
* risk_factors: Patient must have new or continuing risk factors at time of delivery

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 6740ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 2937ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 0 supporting, 0 contradicting, 2 missing evidence items in 2103ms.
[EVALUATION_AGENT] COMPLETED: No supporting evidence found. 2 required evidence categories are missing.
[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 0 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 2
Criteria NOT_SATISFIED : 5
Criteria UNKNOWN : 4
Evidence Fusion Result : EXCLUDED

FINAL DECISION

[PEND] Decision : PEND
Reason : The submitted procedure is explicitly excluded by an applicable National Coverage Determination (NCD). The request is pended for manual review. [NCD_EXCLUDES_PROCEDURE]
Reason Codes: NCD_EXCLUDES_PROCEDURE

Decision Basis:
The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.
* NCD-352-HCPCS: NOT_SATISFIED
* NCD-361-HCPCS: NOT_SATISFIED
* NCD-365-HCPCS: NOT_SATISFIED
* NCD-369-HCPCS: NOT_SATISFIED
* NCD-361-Cfcee62: SATISFIED
* NCD-361-C00c478: SATISFIED
* NCD-361-C634c41: UNKNOWN
* NCD-361-Ceb1ff3: UNKNOWN
* NCD-369-Ca67265: NOT_SATISFIED
* NCD-369-C3acd4a: UNKNOWN
* NCD-369-C9731ee: UNKNOWN
Evidence Fusion: EXCLUDED
DecisionEngine: EXCLUDED -> PEND

[94m2026-08-16 13:19:58 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "Breast reconstruction following mastectomy is required to be covered.", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "procedure_and_diagnosis", [0m
[36m "description": "Patient must have undergone mastectomy (procedure code 11952) with a diagnosis of breast cancer (diagnosis code C50.919)." [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 513 | output_tokens: 84 | total_tokens: 597 | latency: 1398ms [0m

[94m2026-08-16 13:20:01 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m


```
[36m { [0m
[36m   "supporting_evidence": [ [0m
[36m     "Patient with breast cancer undergoing breast reconstruction following radical mastectomy.", [0m
[36m     "Surgical pathology confirmed invasive ductal carcinoma." [0m
[36m   ], [0m
[36m   "contradicting_evidence": [], [0m
[36m   "missing_evidence": [] [0m
[36m } [0m
[93m [USAGE] input_tokens: 749 | output_tokens: 55 | total_tokens: 804 | latency: 1094ms [0m
```

```
[94m2026-08-16 13:20:03 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m   "result": "SATISFIED", [0m
[36m   "evidence_cited": [ [0m
[36m     "Patient with breast cancer undergoing breast reconstruction following radical mastectomy.", [0m
[36m     "Surgical pathology confirmed invasive ductal carcinoma." [0m
[36m   ], [0m
[36m   "explanation": "The patient underwent mastectomy for invasive ductal carcinoma (a subtype of breast cancer), satisfying the
required procedure and diagnosis codes implicitly, as radical mastectomy is a form of mastectomy and invasive ductal carcinoma falls
under C50.919." [0m
[36m } [0m
[93m [USAGE] input_tokens: 364 | output_tokens: 101 | total_tokens: 465 | latency: 1605ms [0m
```

```
[94m2026-08-16 13:20:06 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m   "requirement": "Breast reconstruction is covered only if it follows a medically necessary mastectomy and is not for cosmetic
reasons.", [0m
[36m   "required_evidence": [ [0m
[36m     { [0m
[36m       "category": "medically_necessary_mastectomy", [0m
[36m       "description": "Evidence that the mastectomy was performed for a medical reason (not cosmetic)" [0m
[36m     }, [0m
[36m     { [0m
[36m       "category": "noncosmetic_purpose", [0m
[36m       "description": "Evidence that the breast reconstruction is not performed for cosmetic reasons" [0m
[36m     } [0m
[36m   ] [0m
[36m } [0m
[93m [USAGE] input_tokens: 673 | output_tokens: 108 | total_tokens: 781 | latency: 1562ms [0m
```

```
[94m2026-08-16 13:20:08 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m   "supporting_evidence": [ [0m
[36m     "Patient with breast cancer undergoing breast reconstruction following radical mastectomy", [0m
[36m     "Surgical pathology confirmed invasive ductal carcinoma" [0m
[36m   ], [0m
[36m   "contradicting_evidence": [], [0m
[36m   "missing_evidence": [ [0m
[36m     "Evidence that the breast reconstruction is not performed for cosmetic reasons" [0m
[36m   ] [0m
```

[36m } [0m
[93m [USAGE] input_tokens: 763 | output_tokens: 71 | total_tokens: 834 | latency: 1063ms [0m

[94m2026-08-16 13:20:10 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m

[36m { [0m
[36m "result": "SATISFIED", [0m
[36m "evidence_cited": [[0m
[36m "Patient with breast cancer undergoing breast reconstruction following radical mastectomy", [0m
[36m "Surgical pathology confirmed invasive ductal carcinoma" [0m
[36m], [0m
[36m "explanation": "The mastectomy was performed for invasive ductal carcinoma, a medically necessary reason, and reconstruction following such a mastectomy is inherently noncosmetic per policy." [0m
[36m } [0m
[93m [USAGE] input_tokens: 467 | output_tokens: 80 | total_tokens: 547 | latency: 902ms [0m

[94m2026-08-16 13:20:14 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m

[36m { [0m
[36m "requirement": "The policy recognizes that breast reconstruction procedures are increasingly used due to changes in breast cancer epidemiology, improved surgical techniques, better prostheses, and greater awareness of psychological adjustment needs after surgery.", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "epidemiological_change", [0m
[36m "description": "Evidence of an apparent increase in incidence of breast cancer" [0m
[36m], [0m
[36m { [0m
[36m "category": "surgical_technique_improvement", [0m
[36m "description": "Evidence of improved surgical skills and techniques for breast reconstruction" [0m
[36m], [0m
[36m { [0m
[36m "category": "prosthesis_development_and_psychological_awareness", [0m
[36m "description": "Evidence of continuing development of better prostheses and increasing physician awareness of postsurgical psychological adjustment" [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 727 | output_tokens: 167 | total_tokens: 894 | latency: 2201ms [0m

[94m2026-08-16 13:20:17 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m

[36m { [0m
[36m "supporting_evidence": [[0m
[36m "Patient with breast cancer undergoing breast reconstruction following radical mastectomy.", [0m
[36m "Surgical pathology confirmed invasive ductal carcinoma." [0m
[36m], [0m
[36m "contradicting_evidence": [], [0m
[36m "missing_evidence": [[0m
[36m "Evidence of an apparent increase in incidence of breast cancer", [0m
[36m "Evidence of improved surgical skills and techniques for breast reconstruction", [0m
[36m "Evidence of continuing development of better prostheses and increasing physician awareness of postsurgical psychological adjustment" [0m

[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 811 | output_tokens: 104 | total_tokens: 915 | latency: 1884ms [0m

[94m2026-08-16 13:20:19 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "UNKNOWN", [0m
[36m "evidence_cited": [[0m
[36m "Patient with breast cancer undergoing breast reconstruction following radical mastectomy.", [0m
[36m "Surgical pathology confirmed invasive ductal carcinoma." [0m
[36m], [0m
[36m "explanation": "The patient evidence confirms a single case of reconstruction after mastectomy but does not address the policy's required categories of epidemiological change, surgical technique improvement, or prosthesis/psychological awareness." [0m
[36m } [0m
[93m [USAGE] input_tokens: 575 | output_tokens: 86 | total_tokens: 661 | latency: 1073ms [0m

APPROVE - Breast Reconstruction covered by NCD 64 [NCD RAG Path]

REQUEST

Procedure Code : 11952
Diagnosis : C50.919
State : TX
Patient Age : 48
Clinical Notes : Patient with breast cancer undergoing breast reconstruction following radical mastectomy. Surgical pathology confirmed invasive ductal carcinoma.

POLICY IDENTIFICATION

NCD : 129
Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD 64
Section : title
Text : Breast Reconstruction Following Mastectomy
Similarity : 0.5906
Source : NCD 64
Section : indications_limitations
Text : Reconstruction of the affected and the contralateral unaffected breast following a medically necessary mastectomy is considered a relatively safe and effective noncosmetic procedure. Accordingly, prog
Similarity : 0.5276
Source : NCD 64
Section : item_service_description
Text : During recent years, there has been a considerable change in the treatment of diseases of the breast such as fibrocystic disease and cancer. While extirpation of the disease remains of primary importa
Similarity : 0.4959

CODE MATCHING

HCPCS 11952 -> NCD 129 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-129-HCPCS
Requirement : The requested procedure must be an applicable service under NCD 129.
Type : STRUCTURED
Result : SATISFIED
Policy Ev. : NCD 129 contains HCPCS 11952 in its covered-procedure list.
Patient Ev. : Submitted HCPCS: 11952
Explanation : Procedure 11952 is listed in NCD 129 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C2 [AGENTIC_QWEN] NCD-64-C93dfcd
Requirement : Breast Reconstruction Following Mastectomy
Type : SEMANTIC
Result : SATISFIED
Policy Ev. : Required: procedure_and_diagnosis: Patient must have undergone mastectomy (procedure code 11952) with a diagnosis of breast cancer (diagnosis code C50.919).
Patient Ev. : Patient with breast cancer undergoing breast reconstruction following radical mastectomy.; Surgical pathology confirmed invasive ductal carcinoma.
Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Breast Reconstruction Following Mastectomy
Policy: NCD 64

Required Evidence:
* procedure_and_diagnosis: Patient must have undergone mastectomy (procedure code 11952) with a diagnosis of breast cancer (diagnosis code C50.919).

Patient Evidence:
* Patient with breast cancer undergoing breast reconstruction following radical mastectomy.
* Surgical pathology confirmed invasive ductal carcinoma.

Qwen Result: SATISFIED
Critic Result: VALIDATED
Final Result: SATISFIED

The submitted clinical documentation satisfies this semantic policy criterion. The agentic evaluation chain VALIDATED the result.

(Agentic pipeline completed in 7578ms)

Agent Trace:
[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 2543ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 0 missing evidence items in 2219ms.
[EVALUATION_AGENT] COMPLETED: Supporting evidence found (2 items). No missing evidence categories. Policy requirement may be satisfied.
[QWEN] COMPLETED: Qwen returned SATISFIED. Evidence cited: 2 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: SATISFIED. All 6 checks passed.

C3 [AGENTIC_QWEN] NCD-64-C1c1615

Requirement : Reconstruction of the affected and the contralateral unaffected breast following a medically necessary mastect...

Type : SEMANTIC

Result : SATISFIED

Policy Ev. : Required: medically_necessary_mastectomy: Evidence that the mastectomy was performed for a medical reason (not cosmetic); Required: noncosmetic_purpose: Evidence that the breast reconstruction is not performed for cosmetic reasons

Patient Ev. : Patient with breast cancer undergoing breast reconstruction following radical mastectomy; Surgical pathology confirmed invasive ductal carcinoma

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Reconstruction of the affected and the contralateral unaffected breast following a medically necessary mastectomy is considered a relatively safe and effective noncosmetic procedure. Accordingly, program payment may be made for breast reconstruction surgery following removal of a breast for any medical reason. Program payment may not be made for breast reconstruction for cosmetic reasons. (Cosmetic surgery is excluded from coverage under §1862(a)(10) of the Act.)

Policy: NCD 64

Required Evidence:

- * medically_necessary_mastectomy: Evidence that the mastectomy was performed for a medical reason (not cosmetic)
- * noncosmetic_purpose: Evidence that the breast reconstruction is not performed for cosmetic reasons

Patient Evidence:

- * Patient with breast cancer undergoing breast reconstruction following radical mastectomy
- * Surgical pathology confirmed invasive ductal carcinoma

Missing Evidence:

- * Evidence that the breast reconstruction is not performed for cosmetic reasons

Qwen Result: SATISFIED

Critic Result: VALIDATED

Final Result: SATISFIED

The submitted clinical documentation satisfies this semantic policy criterion. The agentic evaluation chain VALIDATED the result.

(Agentic pipeline completed in 6965ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 2777ms.

[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 1 missing evidence items in 2165ms.

[EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 1 required evidence categories are missing.

Evidence is insufficient for a definitive conclusion.

[QWEN] COMPLETED: Qwen returned SATISFIED. Evidence cited: 2 items.

[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: SATISFIED. All 6 checks passed.

C4 [AGENTIC_QWEN] NCD-64-C6ab152

Requirement : During recent years, there has been a considerable change in the treatment of diseases of the breast such as f...

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Required: epidemiological_change: Evidence of an apparent increase in incidence of breast cancer; Required: surgical_technique_improvement: Evidence of improved surgical skills and techniques for breast reconstruction

Patient Ev. : Patient with breast cancer undergoing breast reconstruction following radical mastectomy.; Surgical pathology confirmed invasive ductal carcinoma.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: During recent years, there has been a considerable change in the treatment of diseases of the breast such as fibrocystic disease and cancer. While extirpation of the disease remains of primary importance, the quality of life following initial treatment is increasingly recognized as of great concern. The increased use of breast reconstruction procedures is due to several factors: A change in epidemiology of breast cancer, including an apparent increase in incidence; Improved surgical skills and techniques; The continuing development of better prostheses; and Increasing awareness by physicians of the importance of postsurgical psychological adjustment.

Policy: NCD 64

Required Evidence:

- * epidemiological_change: Evidence of an apparent increase in incidence of breast cancer
- * surgical_technique_improvement: Evidence of improved surgical skills and techniques for breast reconstruction
- * prosthesis_development_and_psychological_awareness: Evidence of continuing development of better prostheses and increasing physician awareness of postsurgical psychological adjustment

Patient Evidence:

- * Patient with breast cancer undergoing breast reconstruction following radical mastectomy.
- * Surgical pathology confirmed invasive ductal carcinoma.

Missing Evidence:

- * Evidence of an apparent increase in incidence of breast cancer
- * Evidence of improved surgical skills and techniques for breast reconstruction
- * Evidence of continuing development of better prostheses and increasing physician awareness of postsurgical psychological adjustment

Qwen Result: UNKNOWN

Critic Result: VALIDATED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 8499ms)

Agent Trace:

- [POLICY_AGENT] COMPLETED: Identified 3 required evidence categories in 3340ms.
- [CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 3 missing evidence items in 3065ms.
- [EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 3 required evidence categories are missing. Evidence is insufficient for a definitive conclusion.
- [QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 2 items.
- [CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

EVIDENCE FUSION

Criteria SATISFIED : 3
Criteria NOT_SATISFIED : 0
Criteria UNKNOWN : 1
Evidence Fusion Result : COVERED

FINAL DECISION

- [APPROVE] Decision : APPROVE
Reason : All applicable policy criteria were satisfied. The submitted procedure meets the National Coverage Determination criteria. [NCD_CRITERIA_SATISFIED]

Reason Codes: NCD_CRITERIA_SATISFIED

Decision Basis:

All mandatory policy criteria were satisfied.

* NCD-129-HCPCS: SATISFIED

* NCD-64-C93dfcd: SATISFIED

* NCD-64-C1c1615: SATISFIED

* NCD-64-C6ab152: UNKNOWN

Evidence Fusion: COVERED

DecisionEngine: COVERED -> APPROVE

[94m2026-08-16 13:20:23 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m

[36m { [0m

[36m "requirement": "Challenge ingestion food testing is covered only if it is reasonable and necessary for the individual patient, but is explicitly excluded for diagnosis of rheumatoid arthritis, depression, or respiratory disorders.", [0m

[36m "required_evidence": [[0m

[36m { [0m

[36m "category": "exclusion_confirmation", [0m

[36m "description": "Patient must not be undergoing challenge ingestion food testing for the diagnosis of rheumatoid arthritis, depression, or respiratory disorders." [0m

[36m } [0m

[36m] [0m

[36m } [0m

[93m [USAGE] input_tokens: 693 | output_tokens: 96 | total_tokens: 789 | latency: 1547ms [0m

[94m2026-08-16 13:20:25 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m

[36m { [0m

[36m "supporting_evidence": [[0m

[36m "Patient presenting with recurrent severe systemic adverse food reactions. Oral challenge ingestion food testing requested to confirm diagnosis." [0m

[36m], [0m

[36m "contradicting_evidence": [], [0m

[36m "missing_evidence": [] [0m

[36m } [0m

[93m [USAGE] input_tokens: 762 | output_tokens: 52 | total_tokens: 814 | latency: 1083ms [0m

[94m2026-08-16 13:20:28 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m

[36m { [0m

[36m "result": "SATISFIED", [0m

[36m "evidence_cited": [[0m

[36m "Patient presenting with recurrent severe systemic adverse food reactions. Oral challenge ingestion food testing requested to confirm diagnosis." [0m

[36m], [0m

[36m "explanation": "The testing is requested for systemic adverse food reactions, not for the excluded conditions of rheumatoid arthritis, depression, or respiratory disorders, thus satisfying the exclusion criterion." [0m

[36m } [0m

[93m [USAGE] input_tokens: 465 | output_tokens: 81 | total_tokens: 546 | latency: 1266ms [0m

[94m2026-08-16 13:20:30 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m

[36m { [0m

[36m "requirement": "Food allergy testing and treatment must be evaluated based on the specified procedure and diagnosis.", [0m
[36m "required_evidence": [[0m
[36m { [0m
[36m "category": "procedure_code", [0m
[36m "description": "Procedure code 95052 must be documented." [0m
[36m], [0m
[36m { [0m
[36m "category": "diagnosis_code", [0m
[36m "description": "Diagnosis code T78.1XXA must be documented." [0m
[36m } [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 513 | output_tokens: 95 | total_tokens: 608 | latency: 1197ms [0m

[94m2026-08-16 13:20:33 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "supporting_evidence": [[0m
[36m "Patient presenting with recurrent severe systemic adverse food reactions.", [0m
[36m "Oral challenge ingestion food testing requested to confirm diagnosis." [0m
[36m], [0m
[36m "contradicting_evidence": [], [0m
[36m "missing_evidence": [[0m
[36m "Procedure code 95052 must be documented.", [0m
[36m "Diagnosis code T78.1XXA must be documented." [0m
[36m] [0m
[36m } [0m
[93m [USAGE] input_tokens: 749 | output_tokens: 87 | total_tokens: 836 | latency: 1308ms [0m

[94m2026-08-16 13:20:34 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "UNKNOWN", [0m
[36m "evidence_cited": [[0m
[36m "Patient presenting with recurrent severe systemic adverse food reactions.", [0m
[36m "Oral challenge ingestion food testing requested to confirm diagnosis." [0m
[36m], [0m
[36m "explanation": "The supporting evidence describes clinical presentation and a requested test but does not document the
required procedure code 95052 or diagnosis code T78.1XXA." [0m
[36m } [0m
[93m [USAGE] input_tokens: 388 | output_tokens: 82 | total_tokens: 470 | latency: 775ms [0m

[94m2026-08-16 13:20:37 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "requirement": "Cytotoxic Food Tests are not covered.", [0m
[36m "required_evidence": [] [0m
[36m } [0m
[93m [USAGE] input_tokens: 511 | output_tokens: 25 | total_tokens: 536 | latency: 427ms [0m

[94m2026-08-16 13:20:40 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m

[36m "supporting_evidence": [[0m
[36m "Patient presenting with recurrent severe systemic adverse food reactions.", [0m
[36m "Oral challenge ingestion food testing requested to confirm diagnosis." [0m
[36m], [0m
[36m "contradicting_evidence": [], [0m
[36m "missing_evidence": [] [0m
[36m } [0m
[93m [USAGE] input_tokens: 716 | output_tokens: 55 | total_tokens: 771 | latency: 1029ms [0m

[94m2026-08-16 13:20:42 [INFO] [qwen.qwen3-next-80b-a3b] Model generated prediction: [0m
[36m { [0m
[36m "result": "SATISFIED", [0m
[36m "evidence_cited": [[0m
[36m "Cytotoxic Food Tests" [0m
[36m], [0m
[36m "explanation": "The policy explicitly excludes coverage for Cytotoxic Food Tests, and the patient evidence explicitly mentions
this test, thus satisfying the policy criterion." [0m
[36m } [0m
[93m [USAGE] input_tokens: 326 | output_tokens: 60 | total_tokens: 386 | latency: 935ms [0m

APPROVE - Food Allergy Testing covered by NCD 187 [NCD RAG Path]

REQUEST

Procedure Code : 95052
Diagnosis : T78.1XXA
State : TX
Patient Age : 35
Clinical Notes : Patient presenting with recurrent severe systemic adverse food reactions. Oral challenge ingestion food testing requested to confirm diagnosis.

POLICY IDENTIFICATION

NCD : 161
Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

Source : NCD 187
Section : indications_limitations
Text : This procedure is covered when it is used on an outpatient basis if it is reasonable and necessary for the individual patient.
Challenge ingestion food testing has not been proven to be effective in t
Similarity : 0.4837
Source : NCD 266
Section : title
Text : Food Allergy Testing and Treatment
Similarity : 0.4440
Source : NCD 161
Section : title

Text : Cytotoxic Food Tests
Similarity : 0.4312

CODE MATCHING

HCPCS 95052 -> Article 187 -> MATCHED

POLICY CRITERIA

C1 [SQL] NCD-161-HCPCS

Requirement : The requested procedure must not be explicitly excluded by NCD 161.

Type : STRUCTURED

Result : NOT_SATISFIED

Policy Ev. : NCD 161 explicitly excludes HCPCS 95052.

Patient Ev. : Submitted HCPCS: 95052

Explanation : Procedure 95052 is listed in NCD 161 which EXCLUDES coverage. Criterion NOT_SATISFIED by deterministic SQL check.

C2 [SQL] NCD-187-HCPCS

Requirement : The requested procedure must be an applicable service under NCD 187.

Type : STRUCTURED

Result : SATISFIED

Policy Ev. : NCD 187 contains HCPCS 95052 in its covered-procedure list.

Patient Ev. : Submitted HCPCS: 95052

Explanation : Procedure 95052 is listed in NCD 187 covered HCPCS codes. Criterion SATISFIED by deterministic SQL check.

C3 [AGENTIC_QWEN] NCD-187-C05d1fe

Requirement : This procedure is covered when it is used on an outpatient basis if it is reasonable and necessary for the ind...

Type : SEMANTIC

Result : SATISFIED

Policy Ev. : Required: exclusion_confirmation: Patient must not be undergoing challenge ingestion food testing for the diagnosis of rheumatoid arthritis, depression, or respiratory disorders.

Patient Ev. : Patient presenting with recurrent severe systemic adverse food reactions. Oral challenge ingestion food testing requested to confirm diagnosis.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: This procedure is covered when it is used on an outpatient basis if it is reasonable and necessary for the individual patient. Challenge ingestion food testing has not been proven to be effective in the diagnosis of rheumatoid arthritis, depression, or respiratory disorders. Accordingly, its use in the diagnosis of these conditions is not reasonable and necessary within the meaning of section 1862(a)(1) of the Act, and no program payment is made for this procedure when it is so used.

Policy: NCD 187

Required Evidence:

* exclusion_confirmation: Patient must not be undergoing challenge ingestion food testing for the diagnosis of rheumatoid arthritis, depression, or respiratory disorders.

Patient Evidence:

* Patient presenting with recurrent severe systemic adverse food reactions. Oral challenge ingestion food testing requested to confirm diagnosis.

Qwen Result: SATISFIED

Critic Result: VALIDATED

Final Result: SATISFIED

The submitted clinical documentation satisfies this semantic policy criterion. The agentic evaluation chain VALIDATED the result.

(Agentic pipeline completed in 7374ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 1 required evidence categories in 2759ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 1 supporting, 0 contradicting, 0 missing evidence items in 2287ms.
[EVALUATION_AGENT] COMPLETED: Supporting evidence found (1 items). No missing evidence categories. Policy requirement may be satisfied.
[QWEN] COMPLETED: Qwen returned SATISFIED. Evidence cited: 1 items.
[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: SATISFIED. All 6 checks passed.

C4 [AGENTIC_QWEN] NCD-266-C750995
Requirement : Food Allergy Testing and Treatment
Type : SEMANTIC
Result : UNKNOWN
Policy Ev. : Required: procedure_code: Procedure code 95052 must be documented.; Required: diagnosis_code: Diagnosis code T78.1XXA must be documented.
Patient Ev. : Patient presenting with recurrent severe systemic adverse food reactions.; Oral challenge ingestion food testing requested to confirm diagnosis.
Explanation : =====
AGENTIC SEMANTIC EVALUATION

Criterion: Food Allergy Testing and Treatment
Policy: NCD 266

Required Evidence:
* procedure_code: Procedure code 95052 must be documented.
* diagnosis_code: Diagnosis code T78.1XXA must be documented.

Patient Evidence:
* Patient presenting with recurrent severe systemic adverse food reactions.
* Oral challenge ingestion food testing requested to confirm diagnosis.

Missing Evidence:
* Procedure code 95052 must be documented.
* Diagnosis code T78.1XXA must be documented.

Qwen Result: UNKNOWN
Critic Result: VALIDATED
Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 6838ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 2 required evidence categories in 2421ms.
[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 2 missing evidence items in 2519ms.
[EVALUATION_AGENT] COMPLETED: Partial evidence: 2 supporting items found, but 2 required evidence categories are missing.

Evidence is insufficient for a definitive conclusion.

[QWEN] COMPLETED: Qwen returned UNKNOWN. Evidence cited: 2 items.

[CRITIC_AGENT] COMPLETED: Critic VALIDATED Qwen's result: UNKNOWN. All 6 checks passed.

C5 [AGENTIC_QWEN] NCD-161-Cc27674

Requirement : Cytotoxic Food Tests

Type : SEMANTIC

Result : UNKNOWN

Policy Ev. : Cytotoxic Food Tests

Patient Ev. : Patient presenting with recurrent severe systemic adverse food reactions.; Oral challenge ingestion food testing requested to confirm diagnosis.

Explanation : =====

AGENTIC SEMANTIC EVALUATION

Criterion: Cytotoxic Food Tests

Policy: NCD 161

Required Evidence:

- * (Policy Agent could not identify specific categories)

Patient Evidence:

- * Patient presenting with recurrent severe systemic adverse food reactions.
- * Oral challenge ingestion food testing requested to confirm diagnosis.

Qwen Result: SATISFIED

Critic Result: REJECTED

Final Result: UNKNOWN

Insufficient or inconclusive evidence to determine whether this semantic criterion is satisfied. The result is UNKNOWN. Deterministic rules and EvidenceFusion will determine final coverage.

(Agentic pipeline completed in 7899ms)

Agent Trace:

[POLICY_AGENT] COMPLETED: Identified 0 required evidence categories in 3035ms.

[CLINICAL_EVIDENCE_AGENT] COMPLETED: Found 2 supporting, 0 contradicting, 0 missing evidence items in 2830ms.

[EVALUATION_AGENT] COMPLETED: Supporting evidence found (2 items). No missing evidence categories. Policy requirement may be satisfied.

[QWEN] COMPLETED: Qwen returned SATISFIED. Evidence cited: 1 items.

[CRITIC_AGENT] COMPLETED: Critic REJECTED Qwen's result (SATISFIED). Final result: UNKNOWN. Reasons: Qwen cited 1 evidence item(s) that cannot be verified in the request: ['Cytotoxic Food Tests']. Potential hallucination detected.

EVIDENCE FUSION

Criteria SATISFIED : 2

Criteria NOT_SATISFIED : 1

Criteria UNKNOWN : 2

Evidence Fusion Result : EXCLUDED

FINAL DECISION

[PEND] Decision : PEND

Reason : The submitted procedure is explicitly excluded by an applicable National Coverage Determination (NCD). The request is pended for manual review. [NCD_EXCLUDES_PROCEDURE]

Reason Codes: NCD_EXCLUDES_PROCEDURE

Decision Basis:

The request was pended because one or more mandatory policy criteria were not satisfied or evidence was ambiguous.

* NCD-161-HCPCS: NOT_SATISFIED

* NCD-187-HCPCS: SATISFIED

* NCD-187-C05d1fe: SATISFIED

* NCD-266-C750995: UNKNOWN

* NCD-161-Cc27674: UNKNOWN

Evidence Fusion: EXCLUDED

DecisionEngine: EXCLUDED -> PEND

REQUEST_MORE_INFORMATION - Outside jurisdiction (CA) [LCD Path]

REQUEST

Procedure Code : 64483

Diagnosis : M54.17

State : CA

Patient Age : 63

Clinical Notes : Lumbosacral radiculopathy.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED

Jurisdiction : CA -> MATCHED

LCD : 39240 Epidural Steroid Injections for Pain Management

Article : 58992

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

CODE MATCHING

HCPCS 64483 -> Article 58992 -> NOT_FOUND

ICD-10 M54.17 -> Article 58992 -> NOT_FOUND

POLICY CRITERIA

C1 [SQL] ARTICLE-58992-HCPCS

Requirement : The requested procedure must be an applicable service under the Article.

Type : STRUCTURED

Result : NOT_SATISFIED

Policy Ev. : Article 58992 does not contain HCPCS 64483 in its coverage list.

Patient Ev. : Submitted HCPCS: 64483

Explanation : Evaluated as NOT_SATISFIED by SQL.

EVIDENCE FUSION

Criteria SATISFIED : 0

Criteria NOT_SATISFIED : 1
Criteria UNKNOWN : 0
Evidence Fusion Result : UNKNOWN

FINAL DECISION

[INFO] Decision : REQUEST_MORE_INFORMATION

Reason : The request is missing required information: Diagnosis code 'M54.17' not found in policy code lists; Missing explicitly covered diagnosis codes; Clinical documentation required to verify ambiguous criteria. Additional information must be provided before a coverage determination can be made. [MISSING_REQUIRED_INFORMATION]

Reason Codes: MISSING_REQUIRED_INFORMATION

Decision Basis:

Additional information is required to determine coverage.

* ARTICLE-58992-HCPCS: NOT_SATISFIED

Evidence Fusion: UNKNOWN

DecisionEngine: UNKNOWN -> REQUEST_MORE_INFORMATION

REQUEST_MORE_INFORMATION - No matching policy

REQUEST

Procedure Code : 99999

Diagnosis : M54.16

State : TX

Patient Age : 45

Clinical Notes : Unknown procedure code.

POLICY IDENTIFICATION

NCD : NOT_ADDRESSED

Jurisdiction : TX -> NOT_ADDRESSED

POLICY EVIDENCE

RAG : NOT USED - no semantic criterion identified

EVIDENCE FUSION

Criteria SATISFIED : 0

Criteria NOT_SATISFIED : 0

Criteria UNKNOWN : 0

Evidence Fusion Result : NOT_ADDRESSED

FINAL DECISION

[INFO] Decision : REQUEST_MORE_INFORMATION

Reason : No coverage policy references procedure '99999'.

Reason Codes: POLICY_NOT_FOUND

Decision Basis:

No coverage policy references procedure '99999'. Evidence Fusion: NOT_ADDRESSED. DecisionEngine: NOT_ADDRESSED ->

REQUEST_MORE_INFORMATION.

END OF DEMO
